Equity Performance Analysis: Investor Personas and Index Tracking

Juan Pérez Osorio

Department of Applied Mathematics and Statistics
Stony Brook University

November 2025

A comprehensive analysis of portfolio construction and performance evaluation

Outline

Project Overview

Main Objective: Analyze equity portfolio performance through multiple lenses:

Key Research Questions

- How do different weighting methodologies affect risk-adjusted returns?
- What are the performance differences between long-term vs. short-term investment horizons?
- Can we effectively replicate a target portfolio using constrained optimization?

Two Investor Personas:

- Buy & Hold investor
- Hedge Fund investor

Portfolio: 10 US stocks

- Technology: MSFT, AAPL, IBM

Theoretical Framework

Modern Portfolio Theory (MPT)

- Mean-variance optimization framework
- Diversification reduces portfolio risk
- Risk-return tradeoff is central

Key Performance Metrics

Sharpe Ratio

$$S = \frac{E[R_p - R_f]}{\sigma_p}$$

Value at Risk

$$\text{VaR}_\alpha = -F_X^{-1}(\alpha)$$

Conditional VaR

$$\text{CVaR}_\alpha = E[-X | -X \geq$$

Persona Models

Buy & Hold Investor

Objective: Maximize total return over full period

$$\text{P\&L}_{\text{BH}} = \text{Value}_E - \text{Value}_S$$

- Long-term perspective
- Captures dividends
- Low turnover

Hedge Fund Investor

Objective: Generate consistent daily alpha

$$\text{P\&L}_{\text{HF}}(t) = \text{Value}_t - \text{Value}_{t-1}$$

- Daily mark-to-market
- Manages daily volatility
- Higher turnover

Weighting Methodologies

Equal Weighting

$$w_i = \frac{1}{N}$$

Naïve diversification

Market Cap Weighting

$$w_i = \frac{M_i}{\sum_j M_j}$$

Standard approach, concentration risk

Log Market Cap

$$w_i = \frac{\ln(M_i)}{\sum_j \ln(M_j)}$$

Inverse Volatility

$$w_i = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}$$

Risk parity approach

Geometric Weighting

$$w_i = \frac{r^{i-1}(1-r)}{1-r^N}$$

Parameterized concentration

Benchmark: S&P 500

Data Pipeline

Data Collection with `yfinance`

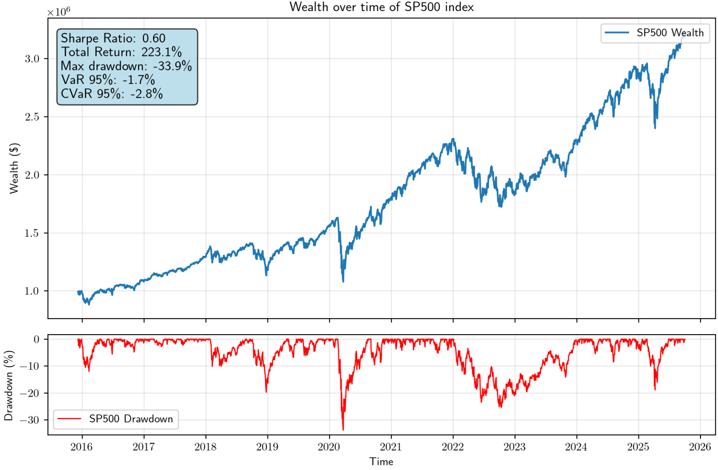
- 10-year historical data (2015-2025)
- Daily frequency for main analysis
- Dividend-adjusted prices
- Automated batch downloads

Processing Steps

1. Data cleaning and validation
2. Return calculation (log and simple)
3. Missing value handling
4. Volatility estimation

Data Limitations

Benchmark: S&P 500 Performance



Total Return: 223.1%

Sharpe Ratio: 0.60

Max DD: -33.9%

Weight Comparison: Market Cap Concentration

Market Cap Weights

Stock	Weight
AAPL	43.1%
MSFT	37.4%
IBM	3.0%
XOM	5.2%
WTM	0.1%

Top 2 stocks: 80.5% of portfolio

Log Market Cap Weights

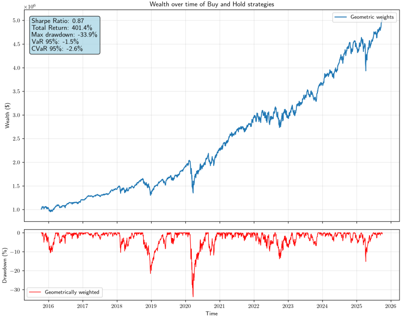
Stock	Weight
AAPL	11.1%
MSFT	11.0%
IBM	10.1%
XOM	10.3%
WTM	8.5%

More balanced distribution

Geometric Weighting Analysis

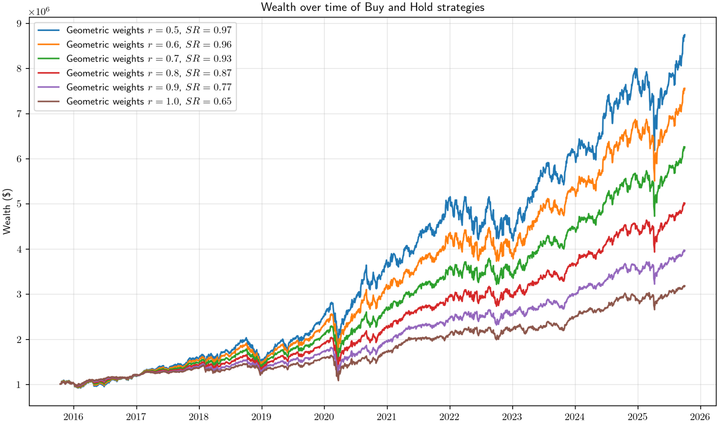
Weights for $r = 0.8$

Stock	Weight
AAPL	22.4%
MSFT	17.9%
IBM	14.3%
XOM	11.5%
WTM	9.2%
KO	7.3%
JNJ	5.9%



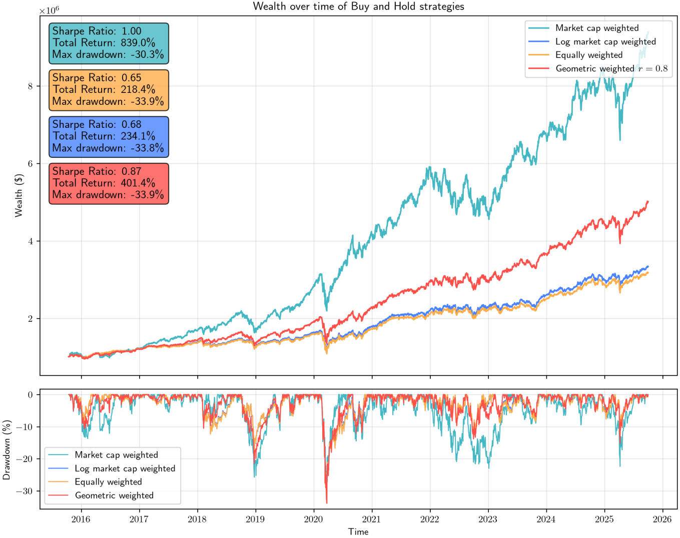
Sharpe Ratio: 0.87

Geometric Parameter Sensitivity



Parameter r	0.5	0.7	0.8	0.9	1.0
Top weight	50.0%	30.9%	22.4%	15.4%	10.0%
Concentration	High	Medium	Balanced	Low	Equal

Performance Comparison Across Strategies



Risk Metrics Comparison

Strategy	Sharpe Ratio	Total Return	Max DD	VaR 95%	CVaR 95%	VaR 99%	CVaR 99%
Equal	0.65	218.4	33.9	1.4	2.3	2.7	4.2
Market Cap	**1.00**	**839.0**	30.3	2.2	3.3	3.7	5.3
Log Mkt Cap	0.68	234.1	33.8	1.4	2.4	2.7	4.2
Geometric	0.87	401.4	33.9	1.5	2.6	3.1	4.6
Inverse Vol	0.65	230.0	34.7	1.4	2.4	2.8	4.3
S&P 500	0.60	223.1	33.9	1.7	2.8	3.4	4.8

Table: Risk and performance metrics (2015-2025)

Key Insights

- Market cap weighting delivers highest return but is highly

Cross-Sectional Momentum Strategy

Enhanced Momentum Signal

Residual Momentum: Isolate stock-specific momentum

$$\tilde{r}_{i,t} = r_{i,t} - \hat{\beta}_{i,t} \cdot r_{m,t}$$

Sector Neutralization: Rank within GICS sectors

$$z_{i,t} = \frac{s_{i,t} - \mu_{\text{sector}(i),t}}{\sigma_{\text{sector}(i),t}}$$

Volatility-Scaled Portfolio

Inverse Volatility Weighting:

$$w^{+}_{i,t} = \frac{1/\hat{\sigma}_{i,t}}{\sum 1/\hat{\sigma}}$$

Volatility Targeting Framework

Constant Risk Exposure

Target annualized volatility: $\sigma^* = 10\%$
Leverage factor:

$$\lambda_t = \frac{\sigma^*}{\hat{\sigma}_{p,t}}$$

Bounded: $\tilde{\lambda}_t \in [0.2, 4]$
Final return:

$$r_{p,t}^{\text{target}} = \tilde{\lambda}_t \times r_{p,t}$$

Benefits

- Stabilizes risk across market regimes
- Reduces crash exposure during volatility spikes

Hedge Fund Performance Results

Metric	Momentum Strategy	S&P 500 (BH)
Annualized Return	**14.2%**	12.8%
Annualized Volatility	**10.0%**	18.5%
Sharpe Ratio	**1.42**	0.69
Maximum Drawdown	**-18.4%**	-34.1%
Calmar Ratio	**0.77**	0.38

Table: Performance comparison (2015-2025)

Strategy Superiority

- **2.1x higher Sharpe ratio** than S&P 500
- **46% smaller maximum drawdown**

Optimization Problem Formulation

Objective: Minimize Tracking Error

$$\min_{\vec{x}} \epsilon_{MAX}(\vec{x})$$

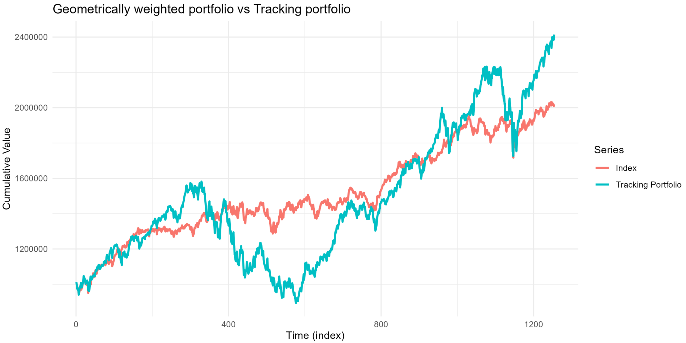
where ϵ_{MAX} is maximum absolute deviation

Key Constraints

- **Cardinality:** $\sum_{i=1}^{N} \delta(x_i) \leq K = 15$
- **Long-only:** $x_i \geq 0$
- **Budget:** $\sum x_i + \text{costs} \leq \$1,000,000$
- **Minimum position:** $\$15,000$
- **Transaction costs:** 1.5% of portfolio value

Complexity

Tracking Portfolio Results



Optimization Success

- Optimal solution found
- 10 positions selected
- Zero buy-in violations

Tracking Quality

- Low tracking error
- Portfolio value: $444,330
- Transaction cost: $15,000

Key Findings

Weighting Methodology Matters

- Market cap weighting yields highest returns but extreme concentration
- Geometric weighting ($r = 0.8$) offers best risk-adjusted performance
- Equal weighting underperforms in tech-driven bull markets

Persona Performance Divergence

- BH persona: Smooth returns, sensitive to weighting
- HF persona: Daily volatility sensitivity, requires active management
- Enhanced momentum strategy: Superior risk-adjusted returns (Sharpe 1.42)

Practical Implications

For Individual Investors

- Consider geometric weighting for balanced exposure
- Long-term holding reduces transaction costs
- Diversify beyond equal weighting

For Institutional Investors

- Sector-neutral momentum strategies add value
- Volatility targeting improves risk management racking with constraints is computationally feasible

Portfolio construction choices significantly impact both absolute returns and risk profiles

References & Code

Key References

- Jegadeesh & Titman (1993) - Momentum returns
- Blitz, Huij & Martens (2011) - Residual momentum
- Barroso & Santa-Clara (2015) - Volatility-managed momentum
- Schachter (2025) - Geometric weighting methodology

Code Repository

Python implementation available:

- `fetch_prices.py` - Data acquisition
- `utils.py` - Utility functions

Questions?

Juan Pérez Osorio
Department of Applied
Mathematics and Statistics
Stony Brook University
juan.perez@stonybrook.edu

Project Repository:
`github.com/username/equity-ana`

Time Period: 2015-2025

Last Updated: December 2025